

**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 2

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Reg No : 22TD0053

Subject Name : E - BUSINESS

Subject Code : CS E63

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Unit : 2

Teacher's Sign :

Department Vision and Mission

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission:

- To impart quality education in computer Science and Engineering by means of learning techniques and value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interaction.
- To build leadership skills, ethical values and teamwork among the students.
- To strengthen collaboration of department and industry through internships, mentorships and professional body activities.

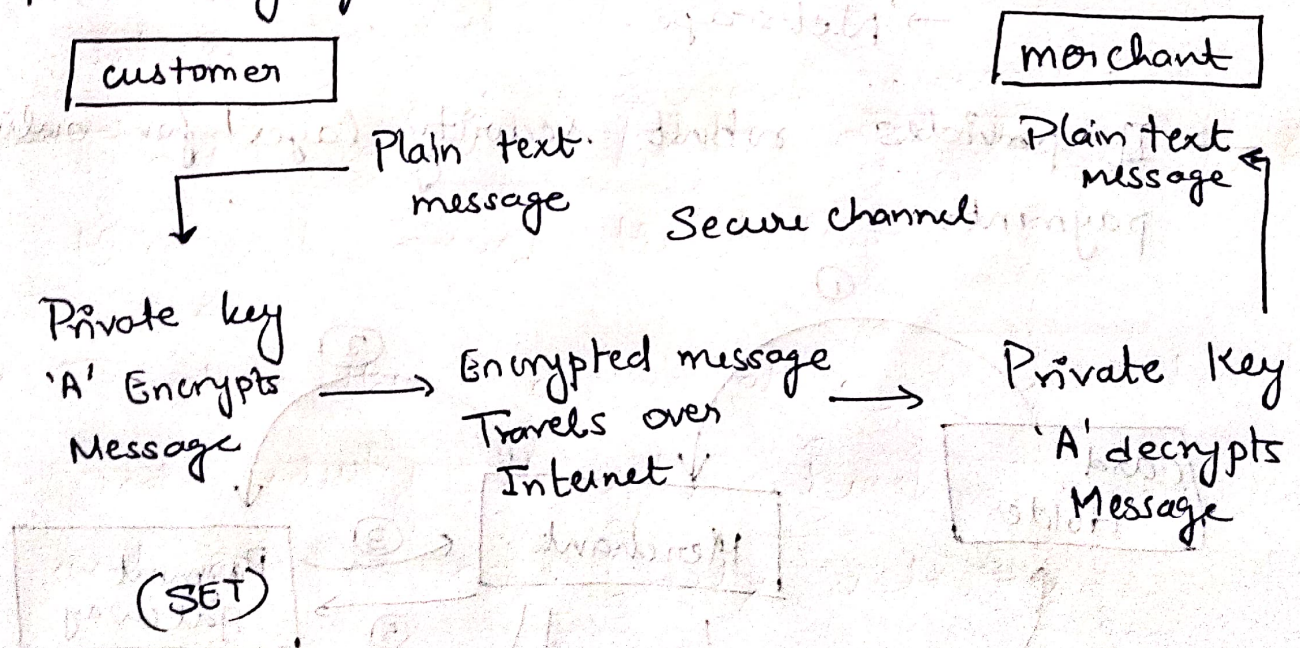
SET

(Secure Electronic Transaction) Protocol

- It is a standard protocol for sending credit card transactions over insecure networks, specifically the internet.
- It is not itself a payment protocol, but rather a set of security protocol and formats that enables users to employ the existing credit card payment infrastructure on an open network in a secure fashion.
- Developed by VISA and MasterCard starting in 1996.

Steps of SET purchase using public key cryptography

Private Key system



Secure purchase using public-key cryptography

- 1) Card Holder requests purchase.
- 2) Merchant contacts payment gateway for Authorization.
- 3) Payment is Authorized.
- 4) Card holder is notified of Authorization.
- 5) Merchant requests payment capture from gateway.
- 6) Token is issued to merchant.
- 7) Merchant redeems token for transfer into its bank account.

→ Initially supported by major organizations

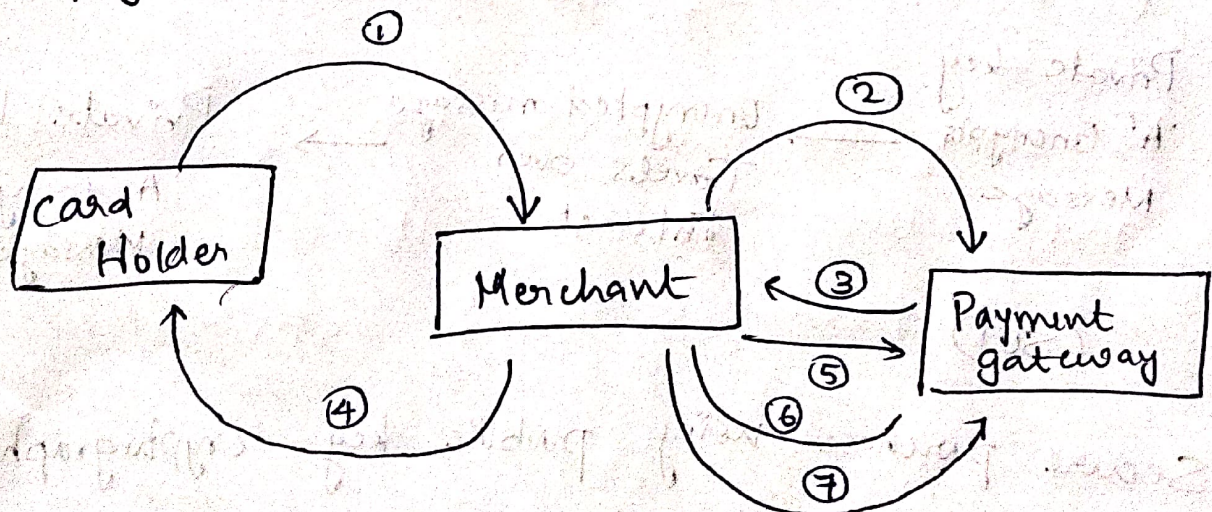
→ Visa

→ Master card

→ Microsoft

→ Netscape.

→ It provides robust security layer for online payments.



→ It is used for credit card Transaction on the web.

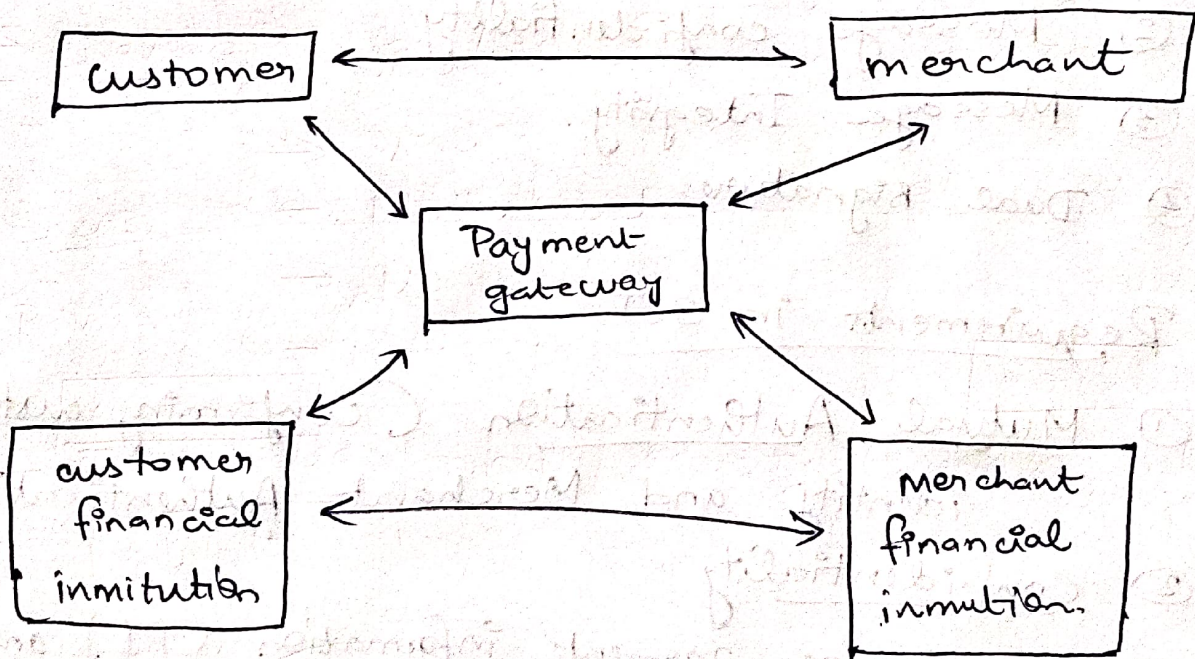
→ It provides

→ Authentication of all parties in Transaction.

→ Confidentiality - a transaction is encrypted to join eavesdroppers.

→ Message integrity - Not possible to alter account Number or transaction account

→ Linkage - Attachments can only be read by third party if necessary.



SET Process work

SET involves the following participants

→ card holder.

The amount making the payment.

→ Issuer - customer's financial institution.

→ Merchant - The seller or service provider

→ Acquirer - The merchant's financial institution.

→ Certificate Authority - An authority issuing standard digital certificates to all participants.

Functionalities of SET

① Authentication

Merchant Authentication

Customer Authentication

② Message confidentiality

③ Message Integrity

④ Dual Signature

Requirements in SET

① Mutual Authentication (confirming customer's identity and Merchant Authentication)

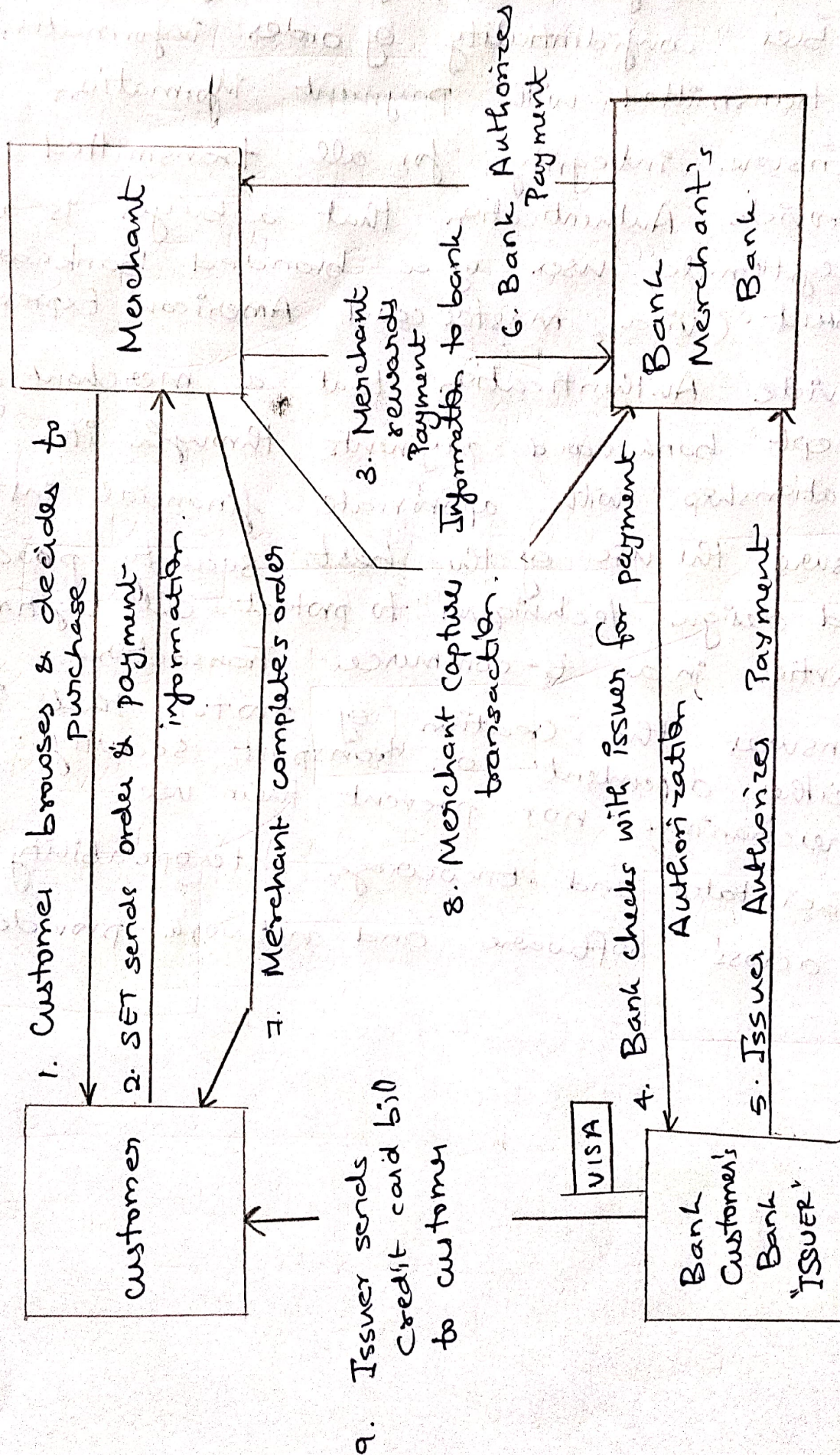
② Confidentiality

Keeping payment information (PI) and order information (OI) confidential through appropriate encryption.

③ Resistance to message modification

④ Interoperability

SET PROCESS



Functions of SET

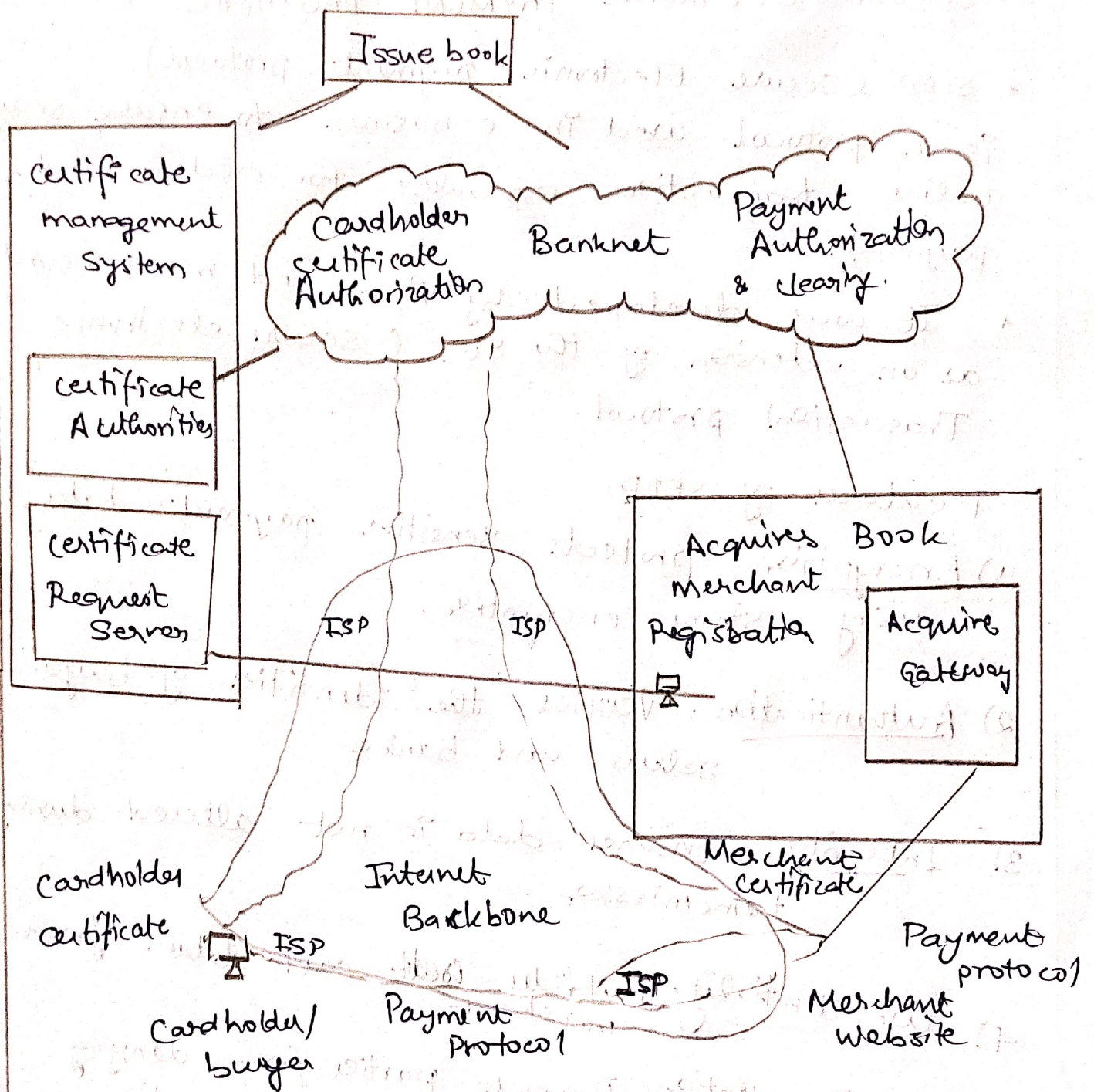
- Provide confidential payment information and enables confidentiality of order information that is transmitted with payment information
- Ensure integrity for all transmitted data.
- Provide Authentication that a buyer is a legitimate user of a branded bankcard account (Visa, master card, American Express)
- Provide Authentication that a merchant can accept bank card payments through its relationship with appropriate financial institution.
- Ensure the use of the best security practises and design techniques to protect all legitimate parties in an E-commerce Transaction.
- Ensures the creation of protocol that is neither dependent on transport security mechanisms nor prevent their use.
- Facilitate and Encourage interoperability across software and network providers

2. SECURE ELECTRONIC PAYMENT PROTOCOL (SEPP)

- SEPP (Secure Electronic payment protocol) is a protocol used in e-business to ensure secure online transactions, especially for credit card payments.
- It was developed by Visa and MasterCard as an extension of the SET (Secure Electronic Transaction) protocol.

Features of SEPP:

- 1) Encryption: protects sensitive payment data using strong encryption.
 - 2) Authentication: verifies the identities of buyers, sellers and banks.
 - 3) Integrity: Ensures data is not altered during transmission.
 - 4) Confidentiality: Hides credit card details from merchants.
 - 5) Non-Repudiation: Prevents parties from denying transaction.
- Network other than Internet. Also possible.



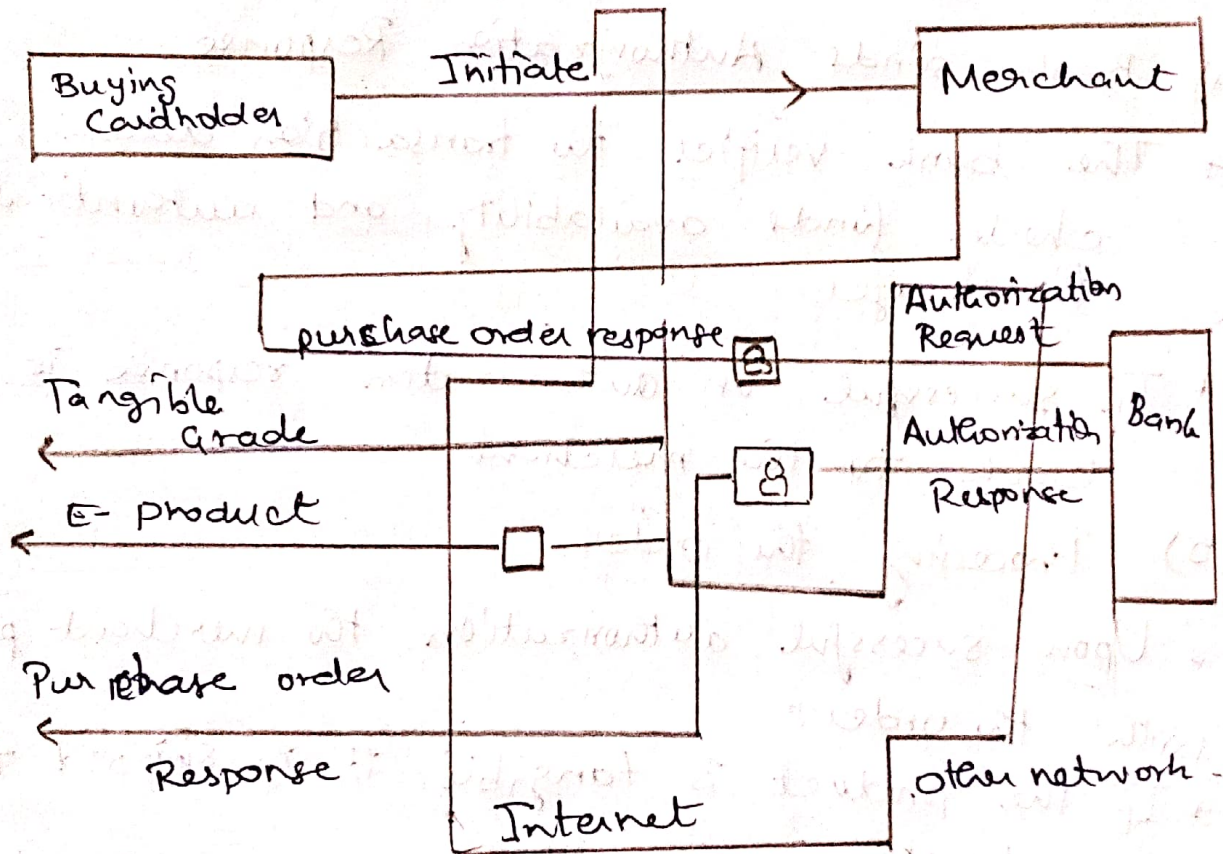
Components of SEPP:

- * Cardholder (Buyer): The customer making an online purchase.
- * Merchant: The online store selling goods or services.
- * Merchant's Bank (Acquirer): Processes the merchant's transactions.
- * Cardholder's Bank (Issuer): Issues credit card to customers.

* Payment Gateway : Acts as an intermediary between the merchant and the bank for processing payments.

SEPP PROCESS:

Time Line



1) Initiation by the buyer (cardholder)

- + The buyer initiates a purchase request on an e-commerce website.
- + The request is sent to the merchant for further processing.

2) Merchant processes purchase order:

- + The merchant receives the purchase order and prepares for payment authorization.
- + A purchase order response is generated and sent back.

3) Authorization Request to Bank:

- + The merchant sends an authorization request to the bank for transaction verification.
- + The request is processed through a secure payment gateway.

4) Bank sends Authorization Response.

- + The bank verifies the transaction details, checks funds availability, and authenticates the buyer.
- + If successful, an authorization response is sent back to the merchant.

5) Processing the order.

- + Upon successful authorization, the merchant proceeds with the order.
- + If the product is tangible, it is shipped to the customer.
- + If the product is an E-product, it is delivered electronically.

6) Final purchase order Response.

- + The transaction is confirmed over the Internet or another secure network.
- + The merchant sends a final response to the buyer, confirming the completion of the order.

Limitations

+ Complex Implementation:

Required software installation for customers and merchants.

+ Slow Adoption:

Businesses preferred simpler alternative like SSL / TLS.

+ High costs

Digital certificate management was expensive.